

20/11/2026

EXPERIMENT NO: 02

Study of different types of network cables

Aim : To study different types of network cables

1) Understand different types of network cable

different types of cables used in networking as

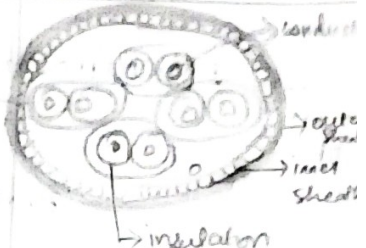
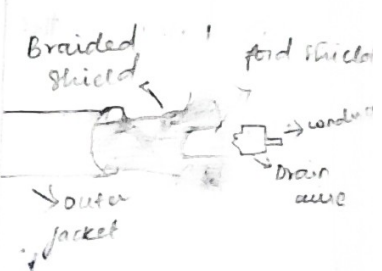
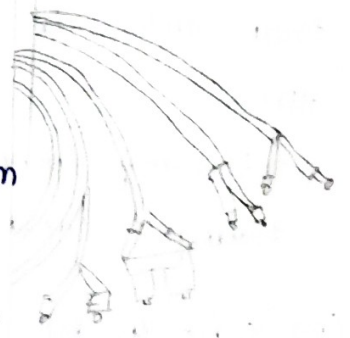
(1) Unshielded Twisted Pair (UTP) cable

(2) Shielded Twisted Pair (STP) cable

(3) Coaxial cable

(4) Fibre optic cable.

Cable type	Category	Maximum data transmission	Advantages & disadvantages	Use	Image
UTP	category 3	10bps	Advantages • cheaper in cost • easy to install as they have a smaller overall diameter	10 Base-T Ethernet	
	category 5	upto 100 Mbps		Fast Ethernet Gigabit Ethernet	
	category 5e	1 Gbps	disadvantages • More prone to EMI & noise		
STP	category 6, 6a	10 Gbps	Advantages • Shielded • Faster than UTP • less susceptible	against ethernet 10G, ethernet 100G widely used in data centers	

SSIP	category	10Gbps	Advantages	Disadvantages	use
			• Expensive • Greater Installation cost.	• Gigabit Ethernet (100m)	
Coaxial Cable	RG-6 RG-9 RG-11	10-100 Mbps	Advantages: • Versatile • High band width • Immune to Interference Disadvantages: • Limited distance • Cost • Size is bulky	Speed of signal is slow Television network High Speed Internet Connection	
Fiber Optics Cable	Single mode Multi mode	100 Gbps	Advantages: • High Speed • High band width • High Security Disadvantages: • Expensive • Skilled installers needed	Maximum distance of fibre optic cable is around 100m	

Result:

Thus the different cables were studied